

Supporting Information for “Aerosols in the vacuum: modelling the time variability of the plume of Enceladus”

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Introduction

This Supporting Information collects the detailed derivations referenced in the main text: the depth-integrated momentum equation that governs the liquid column (Text S1), the energy balance of the thin wall thermal layer that sets the wall sublimation/condensation rate (Text S2), and the quasi-one-dimensional relation that governs the Mach number of the gas column (Text S3).

Text S1. Depth-integrated momentum equation for the liquid column

We use a column coordinate z running from the crack floor ($z = 0$) to the gas–liquid interface ($z = h + L$), where L is the equilibrium water-column depth and h the displacement of the interface from equilibrium. The crack half-gap $\delta(t)$ is prescribed (tidal forcing); a dot denotes d/dt .

Velocity profile from continuity.: For an incompressible column in a slot of uniform gap $\delta(t)$, volume conservation per unit crack length, $\partial\delta/\partial t + \partial(\delta v)/\partial z = 0$, integrates from the floor (where $v = v_0$) to

$$v(z, t) = v_0(t) - \frac{\dot{\delta}}{\delta} z, \quad 0 \leq z \leq h + L, \quad (1)$$

which is Equation (2) of the main text. The velocity at the interface is $v_h \equiv v(h + L, t) = v_0 - (\dot{\delta}/\delta)(h + L)$, giving the kinematic condition $dh/dt = v_h$ (Equation (6) of the main text).

Pointwise momentum equation.: Dividing the momentum balance (Equation (3) of the main text) by $\rho_w \delta$,

$$\frac{\partial v}{\partial t} = -v \frac{\partial v}{\partial z} - \frac{1}{\rho_w} \frac{\partial p}{\partial z} - g - \frac{2C_f}{\delta} v^2 \hat{v}, \quad (2)$$

where the last term is the friction of the two walls (stress $C_f \rho_w v^2$ on each, hydraulic factor $2/\delta$) and $\hat{v} = \text{sgn}(v)$.

Depth integration.: Integrate (2) from $z = 0$ to $z = h + L$. The advective term gives

$$\int_0^{h+L} v \frac{\partial v}{\partial z} dz = \frac{1}{2} (v_h^2 - v_0^2). \quad (3)$$

The pressure term uses the two boundary conditions: a fixed hydrostatic pressure $p(0) = \rho_w g L$ at the floor and $p(h + L) = 0$ at the free surface, so

$$-\frac{1}{\rho_w} \int_0^{h+L} \frac{\partial p}{\partial z} dz = -\frac{1}{\rho_w} [p(h + L) - p(0)] = gL, \quad (4)$$

which combines with the gravity term $-g(h + L)$ to give $-gh$. Writing the friction integral as $F_{\text{fric}} = \int_0^{h+L} (2C_f/\delta) v^2 \hat{v} dz$, the right-hand side of the integrated equation is

$$\int_0^{h+L} \frac{\partial v}{\partial t} dz = -\frac{1}{2} (v_h^2 - v_0^2) - gh - F_{\text{fric}}. \quad (5)$$

Left-hand side.: From (1), the explicit time dependence of the profile is

$$\frac{\partial v}{\partial t} = \frac{dv_0}{dt} - \frac{d}{dt} \left(\frac{\dot{\delta}}{\delta} \right) z = \frac{dv_0}{dt} + \left[\left(\frac{\dot{\delta}}{\delta} \right)^2 - \frac{\ddot{\delta}}{\delta} \right] z, \quad (6)$$

using $\frac{d}{dt}(\dot{\delta}/\delta) = \ddot{\delta}/\delta - (\dot{\delta}/\delta)^2$. Integrating over the column,

$$\int_0^{h+L} \frac{\partial v}{\partial t} dz = (h+L) \frac{dv_0}{dt} + \frac{1}{2} \left[\left(\frac{\dot{\delta}}{\delta} \right)^2 - \frac{\ddot{\delta}}{\delta} \right] (h+L)^2. \quad (7)$$

Result.: Equating (7) and (5) and solving for dv_0/dt ,

$$\frac{dv_0}{dt} = \frac{1}{h+L} \left[-\frac{1}{2}(v_h^2 - v_0^2) - \frac{1}{2} \left[\left(\frac{\dot{\delta}}{\delta} \right)^2 - \frac{\ddot{\delta}}{\delta} \right] (h+L)^2 - gh - \int_0^{h+L} \frac{2C_f}{\delta} v^2 \hat{v} dz \right], \quad (8)$$

which is Equation (4) of the main text. The factor $(h+L)^2$ on the second (unsteady) term follows from the depth integration in (7) and is required dimensionally; the advection term carries the sign $-\frac{1}{2}(v_h^2 - v_0^2)$.

For a height-dependent gap $\delta(z, t)$ (a tapered crack), the same procedure applies with $v(z) = [\delta(z_0)v_0 - \int_{z_0}^z \partial_t \delta dz']/\delta(z)$; the inertial pre-factor $(h+L)$ in (8) is then replaced by $\int_0^{h+L} \delta(z_0)/\delta(z) dz$, and the unsteady term by the corresponding integral of the explicit profile time-dependence.

Text S2. Energy balance of the wall thermal layer

The thin wall layer (region 5, thickness $\lambda \simeq 1$ m, several diurnal skin depths) connects the in-crack heat exchange to the conductive interior. Its surface temperature T_s at a point a distance d below the crack mouth is set by balancing the radiative exchange with the cold sky and the conductive supply from the interior. Treating the surface as grey with the sky at the equilibrium temperature T_e and the conductive flux from the interior (at base temperature T_b) over a path of order d (Equation (8) of the main text, $F_c = 4k[T_s - T_b]/(\pi d)$), the steady balance is

$$2\sigma(T_s^4 - T_e^4) + \frac{4k}{\pi d}(T_s - T_b) = 0, \quad (9)$$

solved for the positive root $T_s(d)$ (a bisection in the code). The net sublimation rate per unit wall area follows from the surface energy released,

$$E(d) = -\frac{\sigma}{L_v} (T_s^4 - T_e^4), \quad (10)$$

with $E < 0$ denoting condensation. Two limits are instructive: near the mouth ($d \rightarrow 0$) the conduction term dominates, $T_s \rightarrow T_b$, and $|E|$ is largest (condensation concentrates within $\sim$ one skin depth of the surface); deep in the crack (d large) the conduction term vanishes, $T_s \rightarrow T_e$, and $E \rightarrow 0$.

When a wall element is instead submerged, it exchanges heat with the liquid by conduction across region 5; the time-dependent diffusion equation $\partial_t T = \kappa \partial_{zz} T$ is integrated in the layer with the sublimation-side boundary set by (10) (when exposed) or by contact with the ~ 273 K liquid (when submerged), and the conduction-side boundary matched to the steady interior profile of Equation (8). The saturated vapour pressure entering the exposed-wall flux uses $p_w = A \exp(-B/T_w)$ with $A = 3.63 \times 10^{12}$ Pa, $B = 6147$ K.

Text S3. Mach-number relation for the gas column

The vapour column (region 2) is modelled as quasi-one-dimensional, compressible, calorically perfect flow in a slot of gap δ , subject to distributed mass addition from wall sublimation (E per unit wall area) and wall friction (C_f). Combining the differential forms of continuity, momentum, energy and the perfect-gas relation for such a flow (the influence-coefficient method), the Mach number M evolves with height z as

$$\frac{dM}{M} \frac{1 - M^2}{1 + \frac{\gamma-1}{2} M^2} = \left(\frac{E}{\rho v \delta} + \frac{\gamma C_f}{2\delta} M^2 \right) dz, \quad (11)$$

which is Equation (9) of the main text. The first source term is mass addition: $E/(\rho v \delta)$ is the fractional rate of mass injection per unit length from the two walls into the column

of cross-section $\propto \delta$. The second is the Fanning wall friction, $(\gamma C_f/2\delta)M^2$, with the factor $1/\delta$ the hydraulic length of the slot. As in adiabatic duct flow, the $(1 - M^2)$ factor makes the two sources drive M toward unity from either side, so the exit Mach number is insensitive to the precise boundary value over the range 1.4–1.8 inferred for the plume; we adopt $M = 1.6$ at the crack mouth.